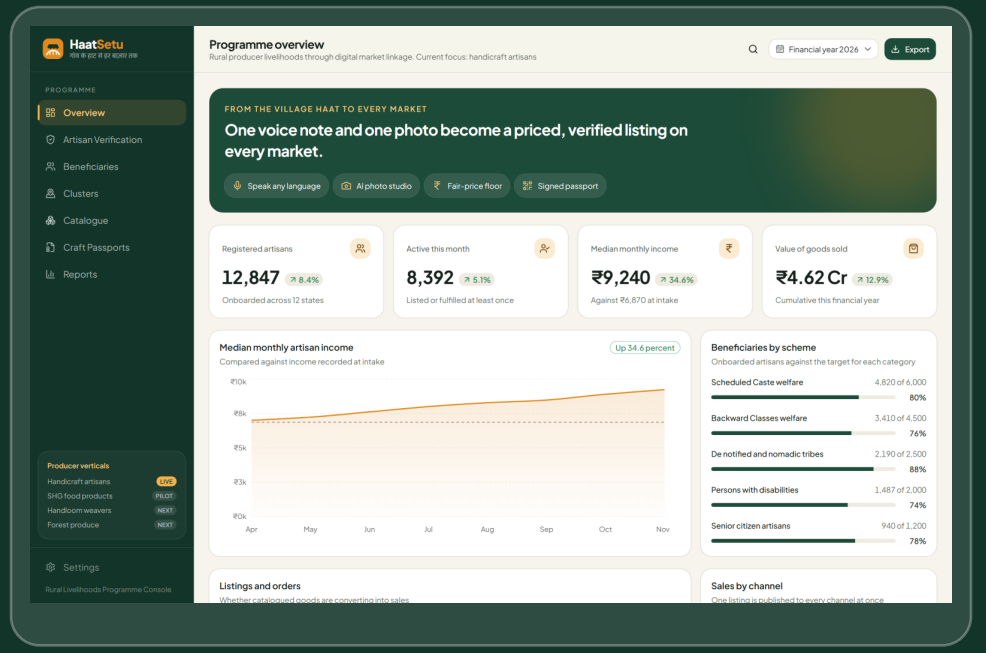
HaatSetu

From the village haat to every market

A voice-first digital market bridge for rural producers



Working prototype: programme console

Theme: Digital Empowerment for Rural Communities

Team Luminescence · [College name]

WHO WE ARE

Team Luminescence



[Member 1]

[Role · e.g. AI & backend]



[Member 2]

[Role · e.g. Mobile app]



[Member 3]

[Role · e.g. Web console]



[Member 4]

[Role · e.g. Research & design]

[College name] · Same-college team of 2–4 members

THE PROBLEM

Rural producers can't reach the market on fair terms

Selling online assumes **literacy, English, product photography, pricing knowledge and marketplace know-how**. Most rural producers have none of these. They depend on the local haat, fairs and middlemen, so income is seasonal and their share of the final price stays small.

We start with handicraft artisans, where the gap between the value of the craft and the maker's income is widest.

64.66 lakh

handloom & handicraft artisans in India, ~64% women

Ministry of Textiles, 2025

₹270 / day

gross value added per craft worker, below minimum wage

IHD–Crafts Council of India

1 : 10

source-to-retail price ratio, "not at all rare"

Dilli Haat study

548 M

rural internet users. Phones are there; selling tools aren't

IAMAI–Kantar, 2025

WHY IT HAPPENS

Four barriers, and they are not unique to crafts

Access

Sales depend on the local haat, seasonal fairs and middlemen who set the price

Language & literacy

Listing means typing English into forms; most producers are comfortable speaking, not typing

Presentation

Cluttered phone photos make a fine product look cheap online

Price & trust

Pricing by guesswork; buyers can't tell genuine handmade or GI goods from copies

The same four barriers hit **10 crore SHG women (90 lakh+ groups)**, handloom weavers and forest-produce collectors. So HaatSetu is built for rural producers in general, starting with crafts.

EXISTING SOLUTIONS

Every option is a place to sell. None creates the listing

Existing solution	What it offers	What's missing
Amazon Karigar, Flipkart Samarth	Marketplace seller programmes	Artisan still shoots photos, writes English, sets price
iTokri, Jaypore	Curated handmade stores	Curator-led and small; not the artisan's own tool
TRIFED / Tribes India, GeM	Government sales channels	One channel each; no listing help
ONDC seller apps	Open-network access	Assume a digitally confident seller
Etsy	Global handmade market	English-first, international, no India context



HaatSetu's gap

Turns one voice note and one phone photo into a finished, fairly priced, verified listing, then publishes it to all of these channels.

OUR SOLUTION

One voice note + one photo → a priced, verified listing

STEP 1 · CAPTURE

Artisan speaks & snaps

- Voice note in own language
- One phone photo
- Materials & hours spent
- Works offline, syncs later

STEP 2 · HAATSETU AI (SERVER-SIDE)

Listing, image, price, proof

Voice → listing
Gemini / Bhashini: bilingual title, story, attributes

AI photo studio
Clean, e-commerce-ready product image

Fair-price engine
Wage floor + LightGBM P10–P90

Craft Passport
Ed25519-signed QR / NFC provenance

STEP 3 · REVIEW

Artisan approves

- Hears listing read back
- Edits by voice or tap
- Price never below the wage floor

STEP 4 · SELL EVERYWHERE

List once, reach all

- HaatSetu storefront
- ONDC & social commerce
- B2B leads, GeM
- Buyer scans QR to verify

Programme console — Artisan verification · Beneficiaries by scheme · Cluster performance · Income at intake vs now · Passport audit

Smartphone only

No special hardware; Flutter app for low-end Android

Works offline

Drafts save on the phone and sync without duplicates

Artisan stays in control

Hears and edits every listing before it goes live

TOP 3 NOVEL FEATURES

What no existing platform does

01

Speak, don't type

A voice note in the artisan's own language becomes a bilingual listing: title, story and attributes. Each language is generated natively, not machine-translated. Gemini, with Bhashini as fallback.

02

Fair-price floor

Price = materials + hours × fair wage + overhead, combined with a market band (P10–P90). The system never suggests a price below the maker's wage, and the server enforces this rule.

03

Signed Craft Passport

A QR/NFC tag with an Ed25519 signature proving artisan, region and technique. Buyers verify it, not just read a claim, which builds trust and supports GI value.

Built on proven, India-first technology

Artisan app	Flutter · offline drafts · voice-first UI · language onboarding
AI services	Gemini audio→listing · Bhashini (MeitY) / Sarvam fallback · AI image studio
Intelligence	LightGBM quantile pricing (P10/P50/P90) · CLIP search in pgvector
Trust & data	FastAPI · PostgreSQL + pgvector on Supabase · Ed25519 passports
Programme console	Next.js dashboard: verification, schemes, clusters, income lift

70 / 70

backend tests pass: pricing rules, passport tampering, degraded modes

Live

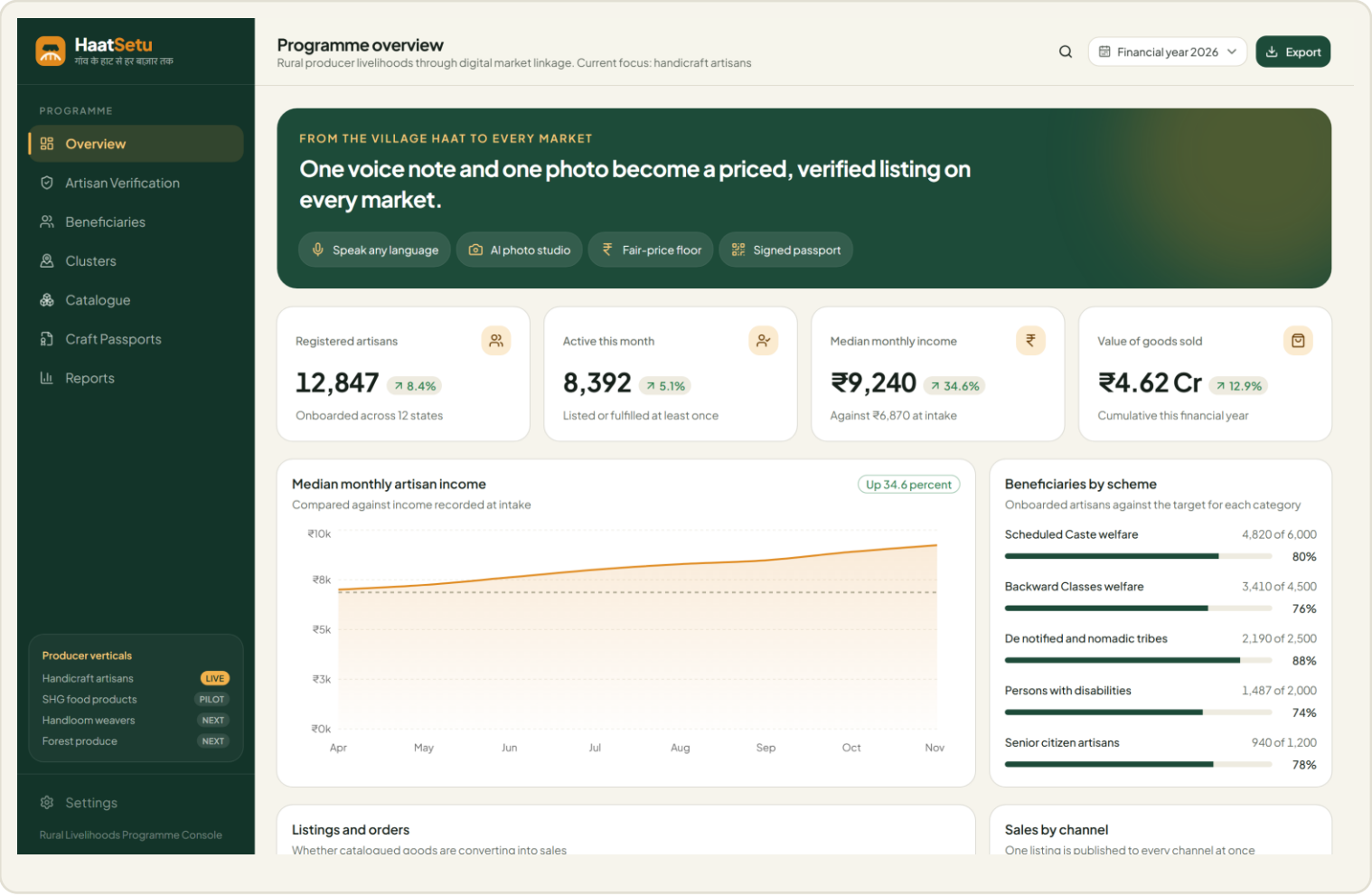
programme console prototype built and running

0

special hardware: a basic smartphone is enough

PROTOTYPE

The programme console is already working



Beneficiaries by scheme

Coverage against target for each welfare category

Income lift

Median income now vs at intake, the measure ministries care about

Artisan verification

Review queue for ID documents before a seller goes live

Producer verticals

Artisans live; SHG food, handloom, forest produce next

Screenshot figures are sample data.

POTENTIAL IMPACT

Better incomes, trusted crafts, visible impact



Producers

- Year-round market access
- From 10+ steps to one input
- Fair-wage price floor



Buyers

- Verified authentic crafts
- Maker's story and region
- Discovery by state & culture



Government

- Income lift measured per beneficiary
- Plugs into GeM, ONDC, Pehchan
- Cluster-level insight



Society & culture

- Digital preservation of crafts
- Women-led livelihoods (~64% women)
- Less migration, local value

IMPLEMENTATION PLAN

Focused today, built to generalise

PHASE 1

Artisans

Pilot with one craft cluster: voice listing, photo studio, fair price, passport

PHASE 2

SHG food products

DAY-NRLM SHGs: pickles, millets, honey. Same pipeline, new categories and "Origin Passport"

PHASE 3

Handloom & forest produce

Weavers and tribal collectors; scale across ONDC, GeM and B2B buyers

Why it scales: voice-to-listing, photo studio, price floor and provenance work for any rural product. A new vertical is a configuration change (vocabulary, categories, price features), not a rewrite.

HaatSetu

गाँव के हाट से हर बाज़ार तक

Speak in your language. Sell to all of India.

Thank you · Team Luminescence · [College name]